

4	(a)		Direction of girl's velocity is changing Acceleration is the rate of change of velocity	
	(b)	(i)	$F = \frac{mv^2}{r} = \frac{(60)(10^2)}{3.4}$ $= 1760 \text{ N}$	
		(ii)	N = centripetal force = 1760 N $f = mg = (60)(9.81)$ $\mu = 0.33$	
	(c)		Vertical component of the normal force is directed upwards it can balance the weight of the girl hence no need for friction	